

Nova, fully by voice: the plan

Plan in phases · 28 September 2026 · general use, for a software engineer · no code yet · research in *Nova_Native_Computer_Use_Research.pdf*

The principle:

1. Make everyday commands instant first.
2. Then handle whatever is on screen.
3. Bring in heavier machinery only where the logs show it's needed.

What drives it: Reflex decides on the Mac in milliseconds. Nova's own hands and Nova Eyes act. Claude no longer drives the screen.

How progress is judged: every phase ends with measured numbers, not "it feels faster". A hearing track runs alongside, because with full voice control a misheard word becomes an action.

Where coding goes: writing and changing code stays with Nova's agents (Claude Code, Codex, OpenCode) in your project folders. Computer use gets you around the Mac. The agents do the coding.

Principles

- **Fastest first.** Every request starts at the top of the ladder and goes down only as far as it must:
 - a direct command (a fixed plan, no looking);
 - an app adapter (the app's own interface);
 - an on-screen step by name;
 - the hard middle;
 - asking you.
- **On the Mac by default.** Nothing about your screen leaves the Mac until Phase 5 decides otherwise, and only if you agree.
- **Built from what already works.** Nova Eyes, the hands skills, Reflex, the trust rules, the activity record and undo, and your projects (Nova already knows their folders).
- **Configurable, not tied to one vendor.** Chrome and Safari; VS Code and JetBrains IDEs; Terminal first, others as options.
- **Each phase ships in small PRs, live one at a time.** Nothing waits for the whole plan.

Phase 0 · Groundwork

Size: small Needs: nothing from you

- **The catalogue** (Appendix A). About 70 general tasks for a software engineer: browser, GitHub, the editor, the terminal, communication, the system and files, typing. It's both the build list and the test set.
- **Baseline.** Time and outcome for each catalogue task today, from the end of speech to done.
- **A log in Nova.** Every command records:
 - how long it took;
 - which rung handled it;

- whether it worked;
- and, when Nova couldn't handle it directly, why.

From here on, that log drives the plan.

- **Recorded screens.** From the start, a sample of screens and element lists is kept locally (only when you switch it on), so Phase 5 can test choices offline.

Done when: the catalogue's baseline numbers exist, and the log is running.

Phase 1 · Instant browser

Size: medium

Needs: which browser first (Chrome, then Safari, unless you say otherwise); whether "stop" works without the wake word while Nova acts

- **Getting to sites.** "Go to GitHub", "open localhost 3000", "open the Vercel dashboard". Spoken names become addresses, in order:
 - known sites;
 - your bookmarks and history, read on the Mac;
 - otherwise, a web search.
- **GitHub from your projects.** Nova knows your projects and their folders, and so their git remotes. That gives direct addresses without a click:
 - "open the Nova PRs" → the repository's pull requests;
 - "PR 28" → that pull request;
 - "its files" → the files it changed;
 - "the Actions" → the workflow runs.
- **Searching.** "Search for ..." uses the default engine. Site searches work too: "search npm for zod", "look it up on MDN / Stack Overflow / GitHub".
- **Tabs and moving around.** New, close, next or previous tab, "the third tab", "reopen the closed tab", back, forward, reload, hard reload, find on the page, zoom, private window, bookmark, copy the page's link.
- **Developer tools.** "Open the console", "open developer tools".
- **Several steps in one sentence.** "Open Chrome, go to GitHub and open the Nova pull requests" is split into steps, and each waits for the screen to settle.
- **A basic brake.**
 - "Stop" while Nova acts.
 - Touching the mouse or keyboard pauses it.
 - Nova always shows what it's doing ("Opening github.com").

How it works: Reflex classifies each step, code reads the details from your words, and the browser is driven through its own interface (Apple Events) and keyboard shortcuts. No model looks at the screen, and nothing leaves the Mac.

Done when: the Phase 1 catalogue tasks finish under 1 s after you stop talking, plus page loading, with no wrong actions, and "stop" halts within 0.3 s.

Phase 2 · Instant apps, coding tools and typing

Size: medium–large Needs: nothing new

- **The editor or IDE** (VS Code, and JetBrains IDEs such as PhpStorm). Each command goes through the IDE's own launcher, links and shortcuts:
 - open a project;
 - open a file by name;
 - go to a line or a symbol, go to the definition, find usages;
 - search the project;
 - run, debug and stop;
 - show the terminal, hide the sidebar, split the editor;
 - format, comment the line, save all;
 - the command palette.
- **The terminal.**
 - "Open a terminal in the Nova project".
 - "Run npm test": the command is typed and run. Risky commands ask first, as Nova's agents' commands already do.
 - "Stop that" (Ctrl-C), "clear", a new tab.
- **Communication.**
 - Slack: "go to the dev channel", "message Sarah ...", both through its quick switcher.
 - Mail: new mail, through mail links.
 - "What's my next meeting?" and "join the meeting", through the calendar Nova.app already reads.
 - Muting and unmuting in Zoom or Meet.
- **Menus by name, in any app.** "Export as PDF", "show the sidebar". Plus a dialog's buttons: "save", "cancel", "don't save", "replace".
- **System Settings panes by name.** "Open Wi-Fi settings".
- **Screenshots.** Of a window or a region.
- **Finder places.** Downloads, "the latest download", "show this file in Finder".
- **Typing and editing by voice.** "Type ...", spoken punctuation, "new line", "delete that", "select the last word", "undo", "copy that", "paste". Your clipboard is put back afterwards.
- **Acting sooner.** Short, complete commands ("new tab", "pause") don't wait the full end-of-turn time. That saves up to about 0.4 s each.

Done when: the Phase 2 catalogue tasks finish under 1 s, with no wrong actions. Typing by voice holds whole sentences with their punctuation.

Track H · Hearing you right (alongside Phases 1–3)

Size: medium Needs: OK for a five-minute "read these commands" recording

- **Act only when sure it heard right.** The recognizer's own confidence, which Nova currently throws away, reaches the decision. When a name or number is doubtful, Nova checks that one thing ("Figma?"). Clear commands still just happen.

- **Stray words.** A bare "yeah" or "okay" confirms nothing unless Nova has just asked something.
- **Names by sound, including the words of your work.** Apps, sites, projects and contacts are matched by how they sound. So are the open project's file and symbol names, and technical words like npm, kubect!, TypeScript, pull request and localhost. Parakeet's vocabulary boosting, which the FluidAudio version Nova already pins includes, and Apple's contextual words both get these names.
- **The wake word.** Near-misses of "Manuel" are tolerated, within limits.
- **Measure your voice.** The recording session, plus a fair test of Apple's recognizer: accurate mode, the best English variant for you, the dictation recognizer.

Done when: on your recorded commands, no misheard word leads to a wrong action, and project and file names come through reliably.

Phase 3 · A proper brake and clear feedback

Size: small Needs: nothing new

- **Stop partway through a step**, not just between steps: between keystrokes, releasing any held key or mouse button.
- **"Stop" heard faster**, from the draft transcript (every 0.7 s) or a small stop-word detector.
- **Three words.**
 - "Wait" pauses, and "carry on" resumes.
 - "Stop" ends the task and says what was done and what wasn't.
 - "Undo" reverses what can be reversed, and says what can't.
- **On screen:** the current step ("2 of 3: opening Pull requests"), a frame on the target, and a Stop button.

Done when: a stop lands within 0.3 s at any moment, and never leaves a key or mouse button held.

Phase 4 · On-screen steps, Reflex first

Size: large Needs: nothing new

- **"Click X", "open the Y tab", "select Z", in any app or page.** Nova Eyes reads the screen's element list, and code and Reflex match your words to element labels, by meaning and by sound. On GitHub: Files changed, Approve, the failing check. In the IDE: its dialogs. In Slack: its buttons. And forms.
- **Acting safely.** The app's own action first, then Accessibility's press, then the mouse. Right before any click by position, Nova checks the target is still under the pointer.
- **Moving and loading pages.** Nova waits until the screen settles (Accessibility quiet, loading finished, picture still), and scrolls by element, never by distance. It finds the target again after every scroll.
- **Checking each step.** Did what should change actually change? If not, one retry, then ask you.
- **Reading the terminal.** "Did the tests pass?", "what did it say?". The terminal's text is read on the Mac, and an answer that needs explaining goes to a brain as a question, not as a driver.
- **Unsure steps are logged**, with their screens, to decide Phase 5.
- **Retire Claude from these tasks.** `computer_task` stops handing everything to Claude. Claude stays for coding and projects.

Done when: the Phase 4 catalogue tasks succeed at least 90% of the time, with no wrong weighty actions and under 1 s a step, and the size of the "unsure" log is known.

Phase 5 · The hard middle

Size: depends on Phase 4 Needs: what may leave the Mac; Jev or a local model; Claude as a last resort or not

- **What it covers:**
 - unfamiliar pages where nothing matches your words ("download the latest release of FluidAudio");
 - unexpected screens: cookie banners, pop-ups, surprise dialogs;
 - judging "is it done?".
- **Decided from the Phase 4 log and the recorded screens:**
 - If it's a small tail, Nova asks you. No Jev.
 - If it's a big middle, Jev handles just those steps (about 0.5 s each; screen text goes to TypeSafe). Or a local model does: private, it can write text too, but it's a download.
- **Privacy controls:** a setting for what may leave the Mac, and a list of local-only apps (your bank, your password manager).

Done when: catalogue tasks in the middle succeed at least 80% of the time, or are handed to you cleanly, with no wrong weighty actions.

Phase 6 · Routines, learned

Size: medium Needs: nothing new

- A successful task is kept as a routine: which app, which elements by label, and what each step should change. Next time it replays natively, checking every screen, and uses judgement only where the screen differs.
- Named routines for your day:
 - "Start my dev setup": the project in the IDE, a terminal running the dev server, localhost in the browser.
 - "Morning check": the open PRs, Slack, the calendar.

Done when: a routine replays at least 3× faster than the first run, and stops cleanly when the screen doesn't match.

Phase 7 · Reach (only if the logs ask for it)

Size: large Needs: OK to install a browser extension

- **Browser extensions**, for Chrome and then Safari, for pages Accessibility can't see well: the page itself, exact clicks, forms, and knowing when something has loaded or stopped moving.
- **Visual fallback** for apps with no Accessibility (games, canvas apps), reading text and controls from the picture on the Mac.

What still asks first

Nova's permission setting decides the questions:

- **Ask first:** a yes before every change.
- **Do what I ask**, the default: your requests are done, the rest is asked.
- **Don't ask:** nothing waits for a yes.

Under the default, these always keep their question:

- merging and pushing;
- force-anything, and deleting branches or repositories;
- deleting permanently;
- sending to a group or outside your team;
- risky terminal commands (`rm -rf`, `git reset --hard`, force push);
- anything to do with money.

Password fields are never typed into, and text on screen is never taken as an instruction.

Order and decisions

Phase	Size	Decision needed before it
0 · Groundwork	Small	None: the catalogue replaces a personal list
1 · Instant browser	Medium	Browser order; "stop" without the wake word
2 · Instant apps, coding tools, typing	Medium–large	None
H · Hearing (alongside 1–3)	Medium	OK for the five-minute recording
3 · Brake and feedback	Small	None
4 · On-screen steps	Large	None
5 · The hard middle	Depends on 4	What may leave the Mac; Jev or a local model; Claude as a last resort
6 · Routines	Medium	None
7 · Reach	Large	A browser extension

How it's measured

- **Speed:** time from the end of your speech to done, at the median (p50) and the slowest 10% (p90), per catalogue task. Targets:
 - direct commands under 1 s, plus loading;
 - screen steps under 1 s each.
- **Right:** how often the task succeeds.
- **Wrong:** actions you didn't ask for. The target is zero of the weighty kind.

- **Stop:** within 0.3 s of hearing "stop", and nothing left held.
- **Where it went:** which rung handled each task, and how often Nova had to ask.

Appendix A · The catalogue: general use, for a software engineer

This is the build list and the test set. "Phase" says when each task becomes instant or reliable. A few tasks already work today, and they're marked.

What you'd say	Phase	How
Browser		
"Go to GitHub" · "open github dot com"	1	Known site → address
"Open localhost 3000"	1	Address from speech
"Search for swift actor reentrancy"	1	Default search engine
"Search npm for zod" · "look it up on MDN / Stack Overflow"	1	Site search addresses
"New tab" · "close this tab" · "next tab" · "the third tab" · "reopen the closed tab"	1	Shortcuts
"Back" · "forward" · "reload" · "hard reload"	1	Shortcuts
"Find 'timeout' on this page"	1	Find, then type
"Zoom in / out / reset" · "new private window" · "bookmark this"	1	Shortcuts
"Copy this page's link"	1	The browser's interface
"Open the console" · "open developer tools"	1	Shortcuts
"Scroll down" · "to the top" · "to the bottom"	1	Keys
"Click Sign in" · "open the Docs link"	4	On-screen step by name
"Put my email in the email field"	4	On-screen step and typing
"Download the latest release of FluidAudio"	5	The hard middle
GitHub		
"Open the Nova repo"	1	Project → git remote → address
"Open the Nova pull requests / issues / actions"	1	Address from the repository
"Open PR 28" · "show its files"	1	Address
"Open the failing check"	4	On-screen step
"Approve this PR"	4	On-screen step; asks first
"Merge it"	4	On-screen step; always asks
Editor or IDE (VS Code, JetBrains)		
"Open Agentic OS in VS Code / PhpStorm"	2	The IDE's launcher, with the project folder
"Open service.ts"	2	Quick open, then type
"Go to line 120" · "go to the definition" · "find usages"	2	Shortcuts

"Search the project for speechOnly"	2	Find in project, then type
"Run the tests" · "run the app" · "stop it" · "debug this"	2	Run configurations and shortcuts
"Show the terminal" · "hide the sidebar" · "split the editor"	2	Shortcuts or menus
"Format this file" · "comment this line" · "save all"	2	Shortcuts
"Run the command 'reload window'"	2	Command palette, then type
"Close this tab" · "close the other tabs"	2	Menus
Terminal		
"Open a terminal in the Nova project" · "new terminal tab"	2	Terminal at that folder
"Run npm test" · "run git status"	2	Typed and run; risky commands ask
"Stop that" · "clear"	2	Ctrl-C; clear
"Did the tests pass?" · "what did it say?"	4	The terminal's text, read on the Mac
Communication and meetings		
"Open Slack" · "go to the dev channel"	2	Quick switcher
"Message Sarah: running late"	2	Quick switcher, then type; shown before it's sent
"Check my email" · "new email to Sam"	2	Mail; a mail link
"What's my next meeting?"	Today	The calendar, through Nova.app
"Join the meeting"	2	Opens the event's meeting link
"Mute" · "unmute" (Zoom, Meet)	2	Shortcuts
System, windows and files		
"Switch to Chrome" · "back to the editor" · "hide the others"	Today / 2	Hands, made faster and more forgiving
"Editor on the left, Chrome on the right" · "to the other screen"	Today	Window control
"Next desktop" · "show all windows"	2	Shortcuts
"Screenshot this window"	2	Screen capture
"Open Downloads" · "the latest download" · "show this in Finder"	2	Finder
"Rename / move / trash that file"	Today	Files skill
"Volume down" · "brightness up" · "Do Not Disturb" · "lock"	Today	System control
"Open Wi-Fi settings"	2	A Settings link
"Export as PDF" · "save" · "don't save"	2	Menus and dialogs by name
Typing and editing		

"Type fix the flaky test"	2	Types your words
"New line" · "comma" · "period"	2	Spoken punctuation
"Delete that" · "select the last word" · "undo" · "copy that" · "paste"	2	Keys; your clipboard put back
Questions about the screen (answers, not control)		
"What does this error mean?" · "summarize this page"	Today	A look at the screen, and an answer from a brain
Routines		
"Start my dev setup" · "morning check"	6	Learned, replayed natively
Coding work (stays with the agents)		
"Ask Claude to fix the failing test" · "have Codex add a README"	Today	Nova's agents, in your project folders